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Stain normalization methods for histopathology image analysis: A comprehensive review and experimental comparison

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ABSTRACT

The advent of whole slide imaging has brought advanced computer-aided diagnosis via medical imaging and artificial intelligence technologies in digital pathology. The examination of tissue samples through whole slide imaging is commonly used to diagnose cancerous diseases, but the analysis of histopathology images through a decision support system is not always accurate due to variations in color caused by different scanning equipment, staining methods, and tissue reactivity. These variabilities decrease the accuracy of computer-aided diagnosis and affect the diagnosis of pathologists. In this context, an effective stain normalization method has proved as a powerful tool to standardize different color appearances and minimize color variations in histopathology images. This study reviews different stain normalization methods highlighting the main methodologies, contributions, advantages, and limitations of correlated works. The state-of-the-art methods are grouped into four distinct categories. Next, we select ten representative methods from the groups and conduct an experimental comparison to investigate the strengths and weaknesses of different methods and rank them according to selected performance accuracy measures. The quality performances of selected methods are compared in terms of quaternion structure similarity index metric, structural similarity index metric, and Pearson correlation coefficient conducting experiments on three histopathological image datasets. Our findings conclude that the structure-preserving unified transformation-based methods consistently outperform the state-of-the-art methods by improving robustness against variability and reproducibility. The comparative analysis we conducted in this paper will serve as the basis for future research, which will help to refine existing techniques and develop new approaches to address the complexities of stain normalization in complex histopathology images.

1. Introduction

Histopathology image analysis is the gold standard to study cancerous diseases by microscopic analysis of tissue through whole slide imaging. A decision support computer-aided diagnosis can play a significant role in routine clinical work to diagnose and stage cancer [1–4]. Now a days it has become one of the major research topics finding several potential applications with medical imaging and AI (Artificial Intelligence) technologies in digital pathology [2–4]. In fact, it provides ways for diagnosing many of medical conditions and phenomena by alleviating the limitations of cancer diagnosis manually done by pathologists or doctors. Normally, computer-aided diagnosis systems help doctors to study the presence, localization, or grading of disease and make decisions for necessary therapies based on the patient's conditions. These systems are useful for the microscopic analysis of tissue samples using whole slide imaging (digitalization of histological tissue slices). With the development of digital pathology, the histopathological tissue slices are scanned using advanced micro-scanners and stored as a single high-resolution digital file [1]. This digital file called whole slide image implies the scanning sections of tissue mounted on a glass slide with high resolution (normally 20×, 40×, or 60× resolution).

In histopathology, whole slides are prepared to identify cells or tissues that are affected by the disease [1]. The majority of whole slide images are captured with brightfield illumination using stains of contrasting colors. In general, different stains are used to facilitate visualization, as the tissue samples look transparent under a microscopic scanner [5]. This is a standard method (staining process) used to detect and visualize cell and tissue morphology by using dyes and antibodies. Hematoxylin and eosin are widely used routine laboratory stains for differentiating between cellular components. Specifically, hematoxylin

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imparts a purple or blue color appearance to cells and nuclei, while eosin imparts a pinkish hue to eosinophilic structures such as muscle fibers, cytoplasm, and collagen [6–9]. But this process causes color variation that affects the quantitative analysis of histopathology images and reduces the accuracy of different computer-aided systems. The conventional image analysis is greatly affected by staining factors, scanner variability, and imaging artifacts requiring robust image pre-processing and post-processing methods for clinically satisfactory results. The color appearance of digital whole slide images may vary due to several factors resulting from the slide preparation and digitization. These include the use of different scanners, specimen preparations, equipment, stain coloring, staining solutions, concentration, oxidation chemistry, pH levels of solutions, staining duration, and manufacturer reactivity [10].

Staining processes are often subject to variability, which complicates the automation of tissue area analysis across different staining techniques. Variations in biological factors across different tissue types and inconsistencies in staining protocols contribute to high levels of visual appearance variability [11–13]. It is therefore important to study how color variability affects image analysis in computer-aided histopathological diagnostics. The challenge of big data also causes a problem, as digital histopathology images can be very large, containing millions of RGB pixels with complex cellular patterns and contextual biology that carry a wealth of information. While computer-aided diagnosis systems are expected to perform well, they must overcome several computational challenges for real-time applications, particularly with high-resolution images where memory and processing hold-ups can occur. Moreover, the performance of computer-aided systems is highly dependent on labeled data, which is often a challenging and time-consuming task when performed manually by pathologists. To address these challenges, stain normalization techniques can be leveraged to transfer all images to a uniform stain combination, thus facilitating the enhanced performance of such systems [14]. Hence, there is an increasing demand to develop an effective and reliable stain normalization technique capable of removing color variability and addressing the heterogeneity of digitalized slides which is still challenging.

One way to reduce color variations present in whole slide images is to convert RGB into grayscale. While grayscale conversion can lead to a loss of several image information, as grayscale images lack the color depth and important details of RGB images [15–17]. The reaction of tissue always varies in response to stains where converting color to grayscale is effective in the presence of a single stain but not in multiple stains. In general, grayscale conversion of multiple stains leads to less accurate diagnosis due to the loss of color information and artifacts [16]. Therefore, color/stain normalization is an important process that involves adjusting the mean color of one image to another image, with the aim of minimizing color variations between them. In this context, researchers concentrate on color/stain normalization, and several techniques are proposed in the literature to adjust colors of tissue whole slide images [6,18–21]. But each of these techniques has it's advantages and limitations in terms of their effectiveness in preserving tissue structures and preventing color artifacts. Thus, our main contributions of this paper are summarized as follows:

1. A comprehensive review of the principal methodologies, including advantages and limitations of different methods in order to highlight potential new perspectives for the future development of novel techniques.
2. The quantitative comparisons of ten methods selected from different categories based on their strengths and weaknesses are evaluated in terms of several image quality evaluation metrics.
3. Finally, our comparative study examines various techniques with the goal of identifying effective methods for addressing challenges related to stain normalization.

The rest of the paper is organized in several sections. Section 2 introduces the steps of histopathology image sample preparation and digitization. Section 3 describes the materials and methods. Section 4 describes a thorough investigative study of different stain normalization techniques with their strengths and weaknesses. Section 5 shows the quality performance comparison of ten selected methods from different state-of-the-art categories. Section 6 highlights possible directions for further development of the new method with a brief description of the main methodologies, contributions, advantages, and disadvantages. Concluding remarks including comprehensive suggestions to determine a suitable method for pathologists and researchers are presented in Section 7.

2. Tissue preparation and whole slide imaging

The current trends in digital pathology are more focused on computerized systems that involve the digitization of glass slides using specialized software, enabling the examination of tissue sections on computer screen. The preparation of whole slides to preserve tissue sample structures is carried out by maintaining different protocols. Typically, histopathological tissue samples are prepared under some fundamental steps as follows,

2.1. Collection and fixation

The first step of whole slide preparation is tissue sample collection which is a process of removing tissue samples from organisms (alive or postmortem samples). Tissue preservation is important to prevent digestion by bacteria and cellular enzymes [22,23]. For the process of fixation, the tissue is immersed in fixatives to prevent both chemical and physical changes. It is essential to choose the adequate fixative. The immediate changes of the sample are called autolysis, which is an enzymatic digestion of tissue affected by insufficient fixation. This process leads to a decreased affinity of stains to the structures of tissue samples. For the particular affinities of fixative, fixation time might be altered for the appearance of histopathology sample [23].

2.2. Dehydration and clearing

Tissue samples are cut into thin sections and submitted to the processing unit of histopathology to dry by removing water and perfusing with paraffin for enough hardness to smooth microtomy. The tissue samples are dehydrated via a sequence of categorized alcohol bath concentrations (from seventy-five to ninety-five percent). After that, it needs to be cleared to remove alcohol and substituted by xylol via successive baths until samples become enough transparent [24]. Dehydration should be performed adequately because water drops might make the sample sections opaque under the microscope which causes unexpected changes into the staining pattern of tissue structure and microscopic details might be lost.

2.3. Paraffin embedding

The histopathology tissue sample containing paraffin block is obtained by embedding the heated paraffin. This step helps to evaporate the solvent and saturates all the spaces inside the tissue sample [24]. A paraffin block containing tissue sample is obtained through this step.

2.4. Microtomy

The paraffin blocks are gradually sectioned by the microtome (high precision cutting instruments) after removing paraffin excess [25]. Then the section is mounted on a glass slide where the thickness of samples (with a thickness of 3–5 μm) permits to recognize tissue structure and cellular morphology (cytoplasm and nucleus) [23]. There might be some influences in sample appearances due to the sectioning process. Thin samples are better for getting more specific information about various components inside nucleus and chromatin distributions [26].

2.5. Staining

Stains are employed after hydrating tissue and removing paraffin to classify cellular and histopathological components in the pre-staining translucent tissue samples [5]. Different mixtures of stains with different spectral absorption characteristics are used to identify and visualize the components and structures of the tissue. Hematoxylin and eosin are the most common routine stains. In general, hematoxylin specifies as a purple or blue color appearance (e.g. cells, nuclei) and eosin highlights eosinophilic structures (e.g., cytoplasm, collagen, and muscle fibers) as a pinkish hue appearance [9,10].

2.6. Mounting

Mounting helps to cover the slides with a coverslip to protect the tissue sample and support better visualization under the microscopic scanners. This process can produce artifacts to the tissue slides, such as dust, bubbles, and contamination with microorganisms. It is hard to normalize the image with dust in a dark color. Moreover, discoloration resulting from the interaction between tissue and stain is a common matter that can lead to color variations, as exemplified by the presence of faded tissue samples [24].

2.7. Digitalization

Histopathological slides are digitalized with the help of different microscopic scanners. The tissue sample digitalization might impose color variations due to several scanning platforms (e.g. ambient illumination, bulbs, sensor chips etc.) and image acquisition (e.g. image compression, brightness correction etc.) [27]. The color appearance of a histological slide can vary significantly when it is scanned by multiple scanners from different manufacturers [5,28]. Most importantly, image blurring might occur if the tissue sample is not properly aligned with scanner's focal plane. Faded tissue samples can be attributed to both the discoloration that occurs when tissue interacts with stains, as well as potential alterations in the way tissue interacts with stains caused by variations in the storage environment and preservation conditions [28]. The presence of these variations has a significant impact on the performance of different algorithms used for automated image analysis in digital pathology. Therefore, stain normalization utilizing color for the classification of pixels into different stain components is an essential part for developing reliable computer-aided analysis of tissue samples [1,9].

3. Materials and methods

This paper systematically reviews and experimentally compares the performance of different color/stain normalization methods. We compose studies published between 2001 to 2023. The Google Scholar[1] and Scopus[2] electronic databases are used to find the relevant papers by several searching keywords such as "Histopathology Image Analysis", "Stain Normalization", "Stain Estimation", "Color Normalization" and "Standardization". Some papers are found by using references of the articles through aforementioned queries and those publications that cited them. This selection process led to the inclusion of 35 studies from the high-relevance category, forming the exclusive foundation of this paper. More specifically, this study is more focused on Web of Science and Scopus-indexed publications.

[1] https://scholar.google.com
[2] https://www.scopus.com

4. Comprehensive review of stain normalization methods

There is an increasing demand to develop stain normalization techniques to minimize the influence of color variations in whole slide images. In this section, we systematically review state-of-the-art stain normalization techniques and divide them into four categories as followed by Zheng et al. [7]. Finally, a comparative analysis is conducted on the main methodologies, contributions, advantages, and limitations of the relevant works in each category. A brief summary of different categories is given below,

4.1. Histogram transformation based approaches

Histogram transformation based methods standardize the color appearance of histopathological images by transforming the color space of the input image to a reference RGB color space with a known distribution of color values. The reference color space is usually derived from a large collection of images that are representative of the staining protocol or laboratory to be normalized. Reinhard et al. [20] introduced a color transformation-based method that employs statistical analysis to transfer color characteristics from one image to another. This method initially converts the RGB image to the $\imath\alpha\beta$ color space, enabling the transfer of background color from the target image to the source image. Subsequently, the image is converted back from the $\imath\alpha\beta$ color space to the RGB color space, resulting in effective color correction.

Zheng et al. [29] proposed a method to normalize histopathology images in HSV (hue-saturation value) space. This method normalizes global illumination and saturation of images into HSV space with precise adjustment of the saturation and value channels and improves normalization performance. Wang et al. [30] proposed a linear color transform approach for automated segmentation and classification of tumor tissues. This method normalizes the images through transforming the color to a template image and a linear projection in $\imath\alpha\beta$ color space. Janowczyk et al. [31] accomplished stain normalization using sparse AutoEncoders to use in standardization of color distributions. The pixels in whole slide images are first clustered based on the features produced by the sparse AutoEncoders. With the help of a specific histogram projection the pixels belonging to the clusters are transformed to the template image. This method follows a subsequent histogram matching and alignment across clusters with different color channels.

Roy et al. [32] proposed a fuzzy-based modified Reinhard color normalization approach as an improvement over the conventional Reinhard method [20]. This approach addresses the limitations of the conventional method by employing fuzzy logic to control contrast enhancement in the $\imath$ space and color coefficients in the $\alpha\beta$ space. This helps to reduce color variation and ensures a global method that satisfies three color normalization hypotheses including correlation coefficient, global mean color in color space and contrast of the image. This method also estimates the number of pixels associated with a homogeneous white luminance portion, further enhancing the overall normalization process. Vijh et al. [33] presented a method that combines fuzzy logic, enhanced fuzzy illuminant, and modified spectral normalization for addressing inconsistencies in biopsy imaging and color variability. The method introduces an enhanced fuzzy modified Reinhard [20] method for illuminant normalization, which effectively handles variations in lighting conditions. As a result, stain normalization is achieved using fuzzy c-mean saturated weighted hue statistics, allowing for accurate adjustment of color variation. The modified spectral normalization technique employed in this method preserves both the histological information and the structural details present in the source image.

Histogram transformation based approaches barely utilize the staining characteristics of the histopathology images. As a result, the background and the region of interest within tissue samples exhibit irregular and confusing patterns, resulting unavoidable presence of undesired artifacts after normalization [19].

4.2. Separated transformation based approaches

Whole slide images are colored based on independent stains so that some methods are proposed based on multiple transformations [18,19,34]. These techniques classify the pixels as different stains using color deconvolution and a matrix decomposition method [21]. Basically, color deconvolution is a traditional method for separating light-absorbing stains and estimating the color proportions in each pixel of an image by utilizing individual spectral characteristics of different stains. This method follows a calibration-based approach to separate light-absorbing stains by utilizing the principles of the Beer–Lambert law (which describes the relationship between light absorption and the concentration of an absorbing substance) [21,35]. Magee et al. [19] introduced a multi-modal color deconvolution based approach including a multi-modal extension to linear normalization in $\imath\alpha\beta$ color space and normalization in a representation space obtained by stain specific color deconvolution. A probabilistic prior and a Variational Bayesian Gaussian Mixture Model are used to achieve per-pixel stain for a unique transformation. This method helps to solve the problems of linear normalization in $\imath\alpha\beta$ color space. Khan et al. [18] proposed a specific color deconvolution approach based on nonlinear mapping of a source to a target image using color deconvolution. The pixels belong to the same stain are extracted and a staining vector is estimated by analyzing pixel distributions of stains. A non-linear mapping is designed to transfer color of different types of pixels to the template image using staining vector.

Bejnordi et al. [36] introduced a hue saturation density-based approach for standardizing the color of whole slide images. This method involves transforming the RGB color space into chromatic and density components. The chromatic and density distributions of each component (hematoxylin, eosin, and background) in the hue saturation density space are aligned to match their respective distributions. Weighting is then applied to the stain for each pixel, followed by the transformation of the hue saturation density color back into the RGB color space. This method ensures the standardization of colors in histopathology images, enabling improved consistency and accuracy in subsequent analyses.

The separated transformation based methods normalize the pixels of the images through different transformations which may bring the color discontinuity into normalized images. This causes improper structural changes in stained whole slides. It is also required additional computational power to normalize the image through multiple transformations.

4.3. Unified transformation based approaches

The drawback of the above-mentioned approaches is the color discontinuity and the high computational time. In this context, several approaches have been proposed to establish a unified transformation for pixels in histopathology images. Instead of pixel-based stain classification, these techniques offer pixel-based extraction of independent stain components. Basically, these approaches extract independent stain components from the pixels and establish a unified transformation for all pixels by utilizing separated stain information. In this context, color deconvolution [21] method has been considered as an effective stain separation method for histopathology image [37–39]. Different color normalization techniques have been employed where the transformation between source and template images are recognized through color deconvolution and the inverse operation of color deconvolution [6,40]. These color deconvolution-based methods provide more reasonable performance and help to preserve the local contrast in stained whole slide images.

Macenko et al. [6] proposed a fringing technique that converts RGB image to optical density space. This approach involves creating a plane (two-dimensional surface that is defined by three non-collinear points or by a point and a normal vector) corresponding to the largest singular values and determining the corresponding angles. By estimating the maximum and minimum angles based on the percentile α^{th} and $(100-\alpha)^{\text{th}}$, a robust estimation result is obtained. The extreme angle values are then transferred back to the optical density space. This method does not have ambiguity in its solution due to Non-negative Matrix Factorization, ensuring that there are no negative coefficients in the color appearance matrix. Zhou et al. [41] introduced the modification of color deconvolution matrix through optimization of hematoxylin and eosin-stained whole slide images. An optimized hematoxylin decomposition on localized images is estimated with the optimization variable as similar to the color deconvolution matrix. Finally, the objective function enforces the background channel to make the deconvolution result zero. But the normalization performance is limited because the specificity of staining is not counted in the optimization process. However, color deconvolution matrices are not independent variables but can be considered as a stain color appearance matrix.

Thus, the stain color appearance matrix (a transformation matrix that helps to extract stain components by linearly combining the color channels) is more acceptable than color deconvolution matrix (a transformation matrix that allows the extraction of individual stain components) as variables of the normalization methods. Li et al. [28] considered the stain color appearance matrix as a model variable and applied non-negative matrix factorization to solve corresponding problems. Additionally, the sparse constraint is considered in non-negative matrix factorization to arrange the pixels as one type of stain [42–44]. Based on this prior knowledge, Vahadane [42] introduced a sparse non-negative matrix factorization (SNMF) based normalization technique. The capacity of color normalization is limited because the stain weights are processed independently after separating stains using sparse constraints. Besides, the overall separated stain's intensity and proportion are not considered in the stain separation process. This takes the risk of a bias in stain separation so that a large number of pixels might be normalized to match the appearance of a single stain. This might result an imbalance or distortion in the representation of different stains in the whole slide image.

To overcome these issues, an adaptive color deconvolution [7] approach has been proposed to separate stains by taking the stain color appearance matrix as variables with an integrated optimization process. The normalization process is done by estimating a unified transformation for all pixels based on the information of separated stains [42,43]. The overall intensity and proportion of stains are considered and embedded in the adaptive color deconvolution approach which successfully overcomes the stain separation problem and avoids color artifacts. The stain intensity normalization parameters are simultaneously estimated where the stain normalization parameters are calculated using pseudo maximum of the separated stains that increase the capacity of handling different color variances. This approach employed exhibits limitations with respect to preserving all the source information with high computational complexity. To evaluate the color appearance matrix, prior or previous knowledge is necessary.

Furthermore, this method faces challenges in analyzing the color distribution of monochromatic whole slide images, particularly in cases where the immunohistochemical stains do not satisfy the Beer–Lambert law [35]. In this case, Retinex model-based stain normalization algorithm [9] effectively removes unwanted variability in histopathological images without any specific failure rate. This method involves first estimating the illumination factors individually by identifying the maximum intensity of the pixel in each of the three color channels. The Retinex model is then utilized to refine the illumination map, taking advantage of the illumination's structural characteristics. The illumination map is adjusted with Gamma correction within the range of [0,1]. The normalized original image is then obtained by multiplying the optical density space with the color appearance matrix and recombining the hematoxylin and eosin components. This method can ideally remove color artifacts and ensure consistent color appearance in analyzing the color distribution of monochromatic histology images, as demonstrated

by the authors. But if one of the two stains is not present in the image properly during the preliminary stain separation process, the separation may fail. It is also hard to restore the original color after normalization using this technique.

Gupta et al. [45] proposed a geometry-inspired chemical-invariant and tissue-invariant stain normalization approach. This method corrects the illumination variations, stain chemical, and stain quantity variations in a unified framework by exploiting the underlying geometry-inspired color vector space. The benchmarking is done on datasets containing three different cell types and prepared with two different staining chemicals. But the alignment of different origins is required for the illumination corrections. Although there has been partial success in overcoming the quadratic objective function of the convergence in the hematoxylin and eosin components, there are still persisting reconstruction errors. Salvi et al. [8] introduced a stain color adaptive normalization algorithm which employs a clustering and segmentation approach to leverage structure detection while preserving dominant colors. Building upon the singular value decomposition-based approach with pixel-wise transformation, this method effectively enhances the contrast between tissue and background while preserving local structures without altering the color of the tissue and background. The preliminary stain separation step may encounter challenges when the image dimensions are very small or if one of the two stains is absent. Moreover, if a stain is present in only a few pixels, the preliminary stain separation may not accurately detect that specific stain.

Unified transformation-based methods extract independent stain components from each pixel instead of categorizing pixels as distinct stains, and then establish a unified transformation for all pixels based on the separated stain information. This unique process allows these methods to effectively preserve both the brightness and structure of the original image, resulting in fewer artifacts after normalization.

4.4. Generative adversarial network based approaches

A major focus of previous research was color normalization and stain separation within histopathology images. In this context, several methods have been proposed for color normalization based on global color normalization, color normalization after supervised stain separation, and color normalization after unsupervised stain separation. Where the global color normalization involves separating the color and intensity information in space, while other two methods require separating the stains either through a supervised or unsupervised method before normalizing the color. Later generative models [46] such as CycleGAN (Cycle-Consistent Adversarial Networks) [47] and StarGAN (Star Generative Adversarial Networks) have been used for stain normalization. These models are able to learn a mapping between stained images from different sources, such as different labs or staining protocols, by using an adversarial training process that encourages the model to produce images that are indistinguishable from the target domain. In addition, some of these models include additional loss functions that aim to preserve the structural features of the tissue, such as the morphology of cells and the texture of the tissue. This can be achieved by using perceptual loss functions, which measure the difference between the high-level features of the input and output images, or by incorporating additional constraints based on prior knowledge of the tissue structure. But it is required to incorporate color normalization techniques such as Reinhard [20] or Macenko [6] into the generative model to achieve color consistency.

Overall, while there is no single generative model that can fully address all the challenges of stain normalization, several models have been developed that can provide a good balance between staining specificity, structure preservation, and color consistency. BenTaieb et al. [46] proposed a Generative Adversarial Networks (GAN) model for stain transformation that combined a GAN with a task-specific network. Another approach [48] based on GAN used grayscale images as input and a classifier to improve performance, but both approaches depend on labeled data and are not generic. This approach causes the loss of color information due to the use of grayscale images. Zanjani et al. [49] introduced an unsupervised generative model based on InfoGAN [50] that performed many-to-one stain transformations, only requiring labels for the target stain style. This model uses only the brightness channel of the Lab color space as input, which can lead to a loss of information. Shaban et al. [51] proposed an unpaired image-to-image translation model based on CycleGAN [47] that used cycle consistency loss to avoid the need for paired images or task-specific labels. However, their model requires labels for both stain styles and performed one-to-one stain transformation. Nishar et al. [52] utilized a pre-trained convolutional neural network (to calculate content and style losses) and a generative adversarial network to normalize whole slide images. Salehi et al. [53] employed a pix2pix method [54] that is trained on paired patches of histopathological images and their corresponding grayscale images to transfer stain, resulting in a many-to-one stain transfer approach.

Liang et al. [55] used a combination of GAN and a directional statistics-based color similarity index to perform stain transformation. To train the network, they computed a feature-preserving loss using a pre-trained classifier network. Kang et al. [56] proposed a stain normalization network for robustness and computational efficiency using two networks. The main network for stain transformation requires paired images for training and is a fully 1×1 convolutional network. The other network is trained using the method [51] and used to generate paired images. Mahapatra et al. [57] proposed a method that combines a pre-trained segmentation network and CycleGAN to preserve semantic features. The feature maps are merged from the segmentation network with the corresponding layers of the generator in CycleGAN, and a feature map loss is used to compare the generated and real images. Liu et al. [58] investigated the use of CycleGAN and a pathological representation network to improve the model's ability to learn pathological features. This method requires only a few images with task-specific labels.

Stain transfer refers to the process of changing the color style of histopathology images while preserving their structures. The process involves two sets of images: source images that require color style transformation and reference or target images that possess the desired color styles. Models such as stain deconvolution or color matching extract stain vectors or distributions from source and reference images to perform stain transfer. Generally, these models do not consider more than one image as the reference image, assume how many stain vectors or distributions exist and require preprocessing, such as segmentation [46]. In addition, manually designing hand-crafted stain vectors can be problematic as they may not be appropriate for all situations. Instead of learning stain information from individual images, generative models use images from complete target styles. It is important to note that most generative models perform one-to-one stain transformations, which are dependent upon labeling both the source and target domains. Many-to-one stain transformations can be performed with monochrome or grayscale images by using just labels for the target domain but may lose critical color information. GANs do not always preserve the structure of the image; to preserve it, some models use cycle-GAN loss or pre-trained networks that require labels for specific tasks.

Table A.1 provides a summary of different color/stain normalization methods used in histopathology image analysis, emphasizing their working principles, strengths, and limitations.

5. Experimental comparison of selected representative methods

This section represents a comparative analysis of ten color/stain normalization techniques from different categories and their effectiveness in histopathology images. These ten methods are selected from the careful analysis of Table A.1 in the appendix and listed in Table 2. The selection criteria include computational complexity, robustness to artifacts, compatibility with downstream applications, availability, and

Table 1
The detailed implementation parameters and their values.

Parameters	Values
Background intensity	255 for 8 bit data
Optical density levels	≤ 0.03
Hyperparameter, μ	$\mu > 0$
Hyperparameter, α, β, λ	$\alpha = \beta = \lambda = 1$ where, $\alpha > 0$, $\beta > 0$, $\lambda > 0$
Constant, t_1	$t_1 = (0.01 \times 255)^2$
Constant, t_2	$t_2 = (0.03 \times 255)^2$
Constant, t_3	$t_3 = t_2/2$
P_{95}	95th percentile
$u(i)$	Average of the RGB channels
p-value	≤ 0.05
QSSIM, SSIM, and PCC	In the range of [0,1]

implementation simplicity, as well as strengths and weaknesses of the methods. Taking these factors into account, we select the representative methods that best meet our specific requirements and are most suitable for our datasets. For the experimental comparison, several image quality evaluation metrics are used to examine the quantitative comparisons of different methods. In the context of our study, we utilize two proprietary datasets and a publicly accessible dataset to evaluate the efficacy of various methodologies in analyzing differently stained histopathology images. We provide detailed descriptions of the datasets, experimental setup, performance evaluation metrics, and experimental outcomes in the following sections.

5.1. Datasets

The following datasets are used in the performance comparison of different methods;

Dataset 1: This dataset comprises 56 Leica Aperio (.svs) images of myoma tissue, rich of collagen fibers, stained with HE (Hematoxylin and Eosin), at an apparent magnification of 40×. The ethical committee of the Northern Ostrobothnia Hospital District gave a favorable evaluation for the collection of tissues. Myoma tissues were used as a foundation to culture oral tongue squamous carcinoma cells, utilizing previously described methods [59]. The samples were digitized with a Leica Aperio AT2 whole slide scanner for preservation.

Dataset 2: This dataset includes a collection of 50 Hamamatsu (.ndpi) images depicting diagnostic lung adenocarcinoma HE-slides selected from surgical resection samples. The slides were stained with Hematoxylin and Eosin stain and captured at an apparent magnification of 40×. The use of diagnostic images was approved by the scientific committee of Biobank of Eastern Finland.

Dataset 3: This dataset is collected from the open access source provided by Salvi et al. [8] including 270 HE-stained tissue images taken from five organs (adrenal gland, breast, colon, liver, prostate) and three magnifications (10×, 20×, 40×). Each image has a fixed dimension of 1024 × 2048 pixels. The image dataset, along with the manual annotations is available at url: https://data.mendeley.com/datasets/sc878z8pm3/1.

5.2. Experimental setup

In our experiment, the original whole slide image annotations are performed using Aperio ImageScope. The methods described in this study are implemented using Python 3.7.18, while some processing is conducted with MATLAB R2022b. All developments and pipeline processing are conducted on a Linux machine featuring an Intel Core i7-8700MQ Quad-Core processor operating at 3.20 GHz. The default value of different parameters, hyper-parameters, and objective functions are used to implement the selected ten representative methods. Detailed specifications of certain parameters and their corresponding values used in our experiment are listed in Table 1.

5.3. Performance evaluation metrics

The color consistency and intensity statistics across the whole slide image are evaluated using the normalized median intensity (NMI) method [60]. To take advantage of the color consistency provided by normalization, we use the formula for measuring normalized median intensity, as recommended by Basavanhally and Madabhushi [60] in their work. The equation used in the study is as follows:

$$NMI(I_{in}) = \frac{\underset{i \epsilon I_{in}}{Median}\{u(i)\}}{\underset{i \epsilon I_{in}}{P_{95}}\{u(i)\}} \tag{1}$$

where I_{in} refers to the processed whole slide image, $u(i)$ denotes the average value of its RGB channels. The median value is represented by $Median()$, and the 95th percentile is denoted by P_{95}. The effectiveness of color normalization methods is evaluated primarily based on image quality. To assess image quality, various statistical metrics such as the Structural Similarity Index Metric (SSIM) [61], Quaternion Structure Similarity Index Metric (QSSIM) [62], and Pearson Correlation Coefficient (PCC) [63] are commonly used. These metrics are widely used for automatic quality analysis of histopathology images. A brief description of each statistical metric is given below.

The Structure Similarity Index Metric (SSIM) is a method for evaluating image quality that takes into account the inter-pixel dependencies that carry information about the structure of the image. The SSIM is defined as follows,

$$SSIM(a,b) = [l(a,b)]^{\alpha}.[c(a,b)]^{\beta}.[s(a,b)]^{\lambda} \tag{2}$$

where, $l(a,b)$, $s(a,b)$, and $c(a,b)$ indicate three factors including luminance, structure, and contrast. Luminance is the product of illumination and reflectance on the surface of the image, but structure is independent of illumination. The SSIM method satisfies the parameters of symmetry, boundedness, and unique maximum. Human vision is highly sensitive to structural and luminance information in visual scenes, and SSIM compares three functions (luminance, structure, and contrast) between the source and processed images. The SSIM equation is formed by combining equations (3), (4), and (5), and provides a reference metric for measuring differences in structural, luminance, and contrast between the source and processed images. The quality of the processed image is evaluated in relation to the initial or source image, with a numerical SSIM value ranging from 0 to 1 where values closer to 1 indicate better color normalization.

$$Luminance, l(a,b) = \frac{\mu_a \mu_b + t_1}{\mu_a^2 + \mu_b^2 + t_1} \tag{3}$$

$$Contrast, c(a,b) = \frac{2\sigma_{ab} + t_2}{\sigma_a^2 + \sigma_b^2 + t_2} \tag{4}$$

$$Structure, s(a,b) = \frac{\sigma_{ab} + t_3}{\sigma_a \sigma_b + t_3} \tag{5}$$

The variables μ_a and μ_b represent the sample means of the source and processed images, while σ_a and σ_b denote their standard deviations. The correlation coefficient between the source and processed images is denoted by σ_{ab}, and the constants t_1, t_2, and t_3 are used to stabilize the SSIM when it approaches zero. The mathematical formula for the structural similarity index metric is derived by combining equations (3), (4), and (5) and is presented as follows,

$$SSIM(a,b) = \left\{ \left(\frac{\mu_a \mu_b + t_1}{\mu_a^2 + \mu_b^2 + t_1} \right) \left(\frac{2\sigma_{ab} + t_2}{\sigma_a^2 + \sigma_b^2 + t_2} \right) \right\} \tag{6}$$

While SSIM helps to measure scalar cross-correlation, QSSIM helps to measure both scalar and cross-correlation or color cross-correlation using quaternions. Hence, QSSIM is better than SSIM in measuring combined degradation because it measures both size and direction, while SSIM only measures size. QSSIM can measure luminance, chrominance, combined degradation, and single degradation. The numerical value of QSSIM also lies between 0 to 1, with a value closer to 1 indicating a

Table 2
The comparison results conducted by measuring the average Euclidean distance of the hematoxylin and eosin vectors.

Methods	Dataset 1		Dataset 2	
	Average Euclidean distance (Mean ± STD)		Average Euclidean distance (Mean ± STD)	
	Hematoxylin	Eosin	Hematoxylin	Eosin
Reinhard et al. [20]	0.118 ± 0.023	0.109 ± 0.019	0.880 ± 0.021	0.852 ± 0.015
Macenko et al. [6]	0.040 ± 0.011	0.038 ± 0.015	0.021 ± 0.015	0.025 ± 0.010
Khan et al. [18]	0.192 ± 0.027	0.183 ± 0.030	0.673 ± 0.032	0.590 ± 0.016
Bejnordi et al. [36]	0.127 ± 0.092	0.022 ± 0.021	0.543 ± 0.018	0.412 ± 0.013
Vahadane et al. [42]	0.017 ± 0.005	0.033 ± 0.009	0.016 ± 0.003	0.020 ± 0.004
Janowczyk et al. [31]	0.053 ± 0.041	0.097 ± 0.037	0.043 ± 0.022	0.880 ± 0.017
Zheng et al. [29]	0.079 ± 0.022	0.145 ± 0.042	0.053 ± 0.017	0.059 ± 0.028
Zheng et al. [7]	0.036 ± 0.004	0.040 ± 0.007	0.012 ± 0.002	0.016 ± 0.003
Gupta et al. [45]	0.040 ± 0.011	0.038 ± 0.015	0.029 ± 0.014	0.026 ± 0.020
Salvi et al. [8]	0.015 ± 0.002	0.026 ± 0.005	0.023 ± 0.002	0.019 ± 0.004
Hoque et al. [9]	0.010 ± 0.001	0.020 ± 0.004	0.015 ± 0.003	0.021 ± 0.003

Table 3
The comparison results conducted by measuring the Standard Deviation (SD) and Coefficient Variation (CV) of Normalized Median Intensity (NMI).

Methods	Dataset 1				Dataset 2			
	Hematoxylin		Eosin		Hematoxylin		Eosin	
	(NMI) SD	(NMI) CV	(NMI) SD	(NMI) CV	(NMI) SD	(NMI) CV	(NMI) SD	(NMI) CV
Reinhard et al. [20]	0.017	0.29	0.022	0.08	0.041	0.23	0.030	0.09
Macenko et al. [6]	0.011	0.15	0.019	0.04	0.009	0.05	0.017	0.04
Khan et al. [18]	0.025	0.39	0.040	0.07	0.019	0.08	0.014	0.06
Bejnordi et al. [36]	0.040	0.33	0.030	0.10	0.038	0.27	0.033	0.08
Vahadane et al. [42]	0.024	0.10	0.020	0.08	0.025	0.06	0.017	0.05
Janowczyk et al. [31]	0.050	0.21	0.030	0.19	0.043	0.15	0.028	0.11
Zheng et al. [29]	0.064	0.27	0.050	0.20	0.053	0.11	0.033	0.13
Zheng et al. [7]	0.030	0.12	0.020	0.03	0.011	0.02	0.009	0.03
Gupta et al. [45]	0.029	0.12	0.023	0.09	0.030	0.07	0.019	0.05
Salvi et al. [8]	0.014	0.03	0.017	0.03	0.018	0.05	0.012	0.04
Hoque et al. [9]	0.010	0.02	0.015	0.02	0.013	0.02	0.011	0.03

better color normalization method. QSSIM is mathematically defined as follows,

$$QSSIM_{R,N} = \left| \left(\frac{2\mu_{qR}\mu_{qN}}{\mu_{qR}^2 + \mu_{qN}^2} \right) \left(\frac{2\sigma_{qR,qN}}{\sigma_{qR}^2 + \sigma_{qN}^2} \right) \right| \tag{7}$$

The Pearson correlation coefficient measures the linear correlation between two images and its range is from 0 to 1. A value of 0 indicates that there is no similarity between the images, while a value greater than 0 means that there is some correlation between the images. The Pearson correlation coefficient is defined mathematically in Eq. (8).

$$PCC = \frac{\sum_i (a_i - \mu_a)(b_i - \mu_b)}{\sqrt{\sum_i (a_i - \mu_a)^2} \sqrt{\sum_i (b_i - \mu_b)^2}} \tag{8}$$

The variables a_i and b_i refer to the source and normalized images, while μ_a and μ_b represent the means of the source and normalized images. We also calculate the relative square error (rSE) [8] as per the formula given below.

$$rSE = \frac{\left\| (M - M_G)^T (M - M_G) \right\|}{\sqrt{\left\| M^T . M \right\| \left\| (M_G)^T . M_G \right\|}} = \frac{\left\| M - M_G \right\|_F^2}{\left\| M \right\|_F . \left\| M_G \right\|_F} \tag{9}$$

where, M refers to the estimated stain color appearance matrix and F is the Frobenius norm of the matrix as defined by Salvi et at. [8]. The rSE value is computed by comparing the ground truth staining matrix (M_G) with the estimated color appearance matrix (M) obtained after the process of separating the stains. Finally, the standard deviation and coefficient variation of the normalized median intensity is computed for all sample images, where a lower value indicates more consistent normalization.

5.4. Experimental results

Our experimental analysis involves annotating the whole slide image using Aperio ImageScope, with the annotated image being automatically analyzed by different state-of-the-art methods. To establish the ground truth labels, we use ImageJ [64] software, which is a manual stain normalization and segmentation technique. The labeling of pixel characteristics, such as hematoxylin, eosin, and background, is performed based on the selection of regions from the datasets. We investigate stain normalization by utilizing the color information of stained whole-slide images. Initially, we employ staining difference and color normalization techniques on whole slide images stained with hematoxylin and eosin. Subsequently, we evaluate the performance and variability of the results by analyzing the hematoxylin and eosin stain vectors under different color intensity conditions. Our investigation delves into examining objective means of measuring variations in color deconvolution results. To appraise the performance of different methods, comparative evaluations are conducted that involved calculating stain vectors and achieving color constancy. In order to compare the efficiency and robustness of different algorithms, we measure the correct stain vectors using the mean and standard deviation of the Euclidean distances.

Table 2 represents the mean and standard deviation of the Euclidean distance between stain vectors obtained from the annotated image and those derived by each algorithm. A higher mean value and lower standard deviation of the Euclidean distances suggest more efficient and robust outcomes. The results show that Hoque et al. [9], Zheng et al. [7], Salvi et al. [8], and Vahadane et al. [42] perform significantly better than other methods for Dataset 1, with lower standard deviations of the Euclidean distances. Similarly, for Dataset 2, the algorithms by Zheng et al. [7], Hoque et al. [9], Vahadane et al. [42], and Salvi et al. [8] exhibit superior performance with lower standard deviations of the Euclidean distances. Table 3 represents the measurement of normalized median intensity, which is used to evaluate the color constancy of different regions.

In Dataset 1, it is observed that Hoque et al. [9], Zheng et al. [7] and Salvi et al. [8] demonstrate significantly better performance in calculating normalized median intensities for hematoxylin and eosin stains as compared to the other reference methods. Similarly, in Dataset

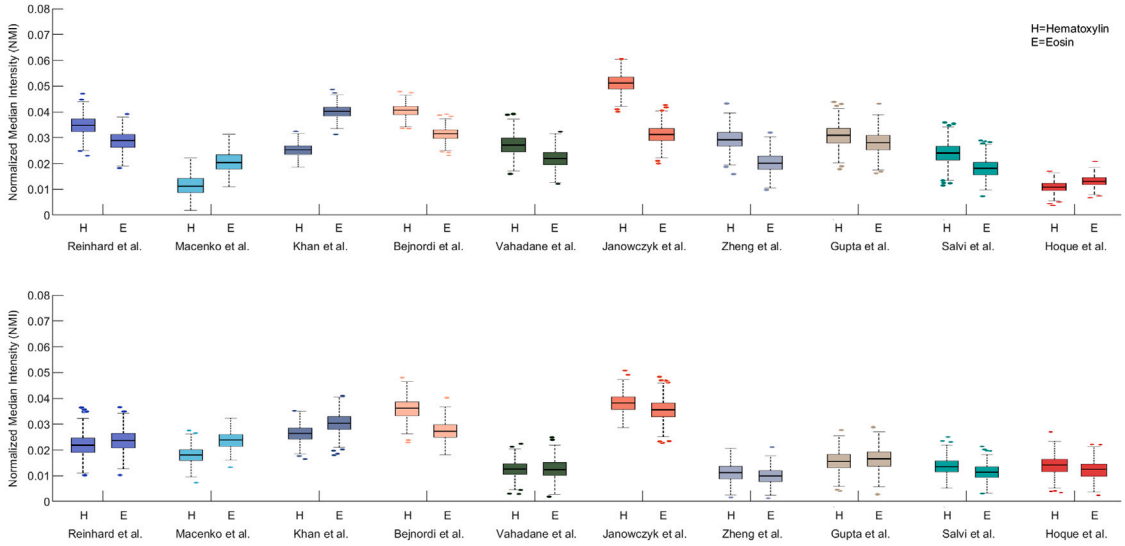


Fig. 1. Boxplots to utilize distribution characteristics including shape, central value, and variability of different state-of-the-art methods by comparing their standard deviations of normalized median intensities.

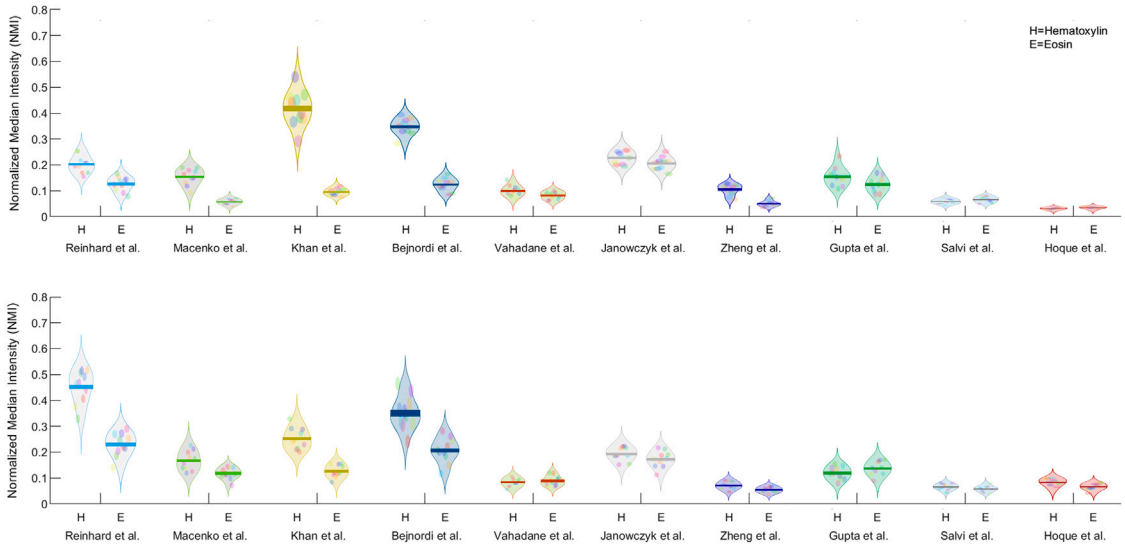


Fig. 2. Violin plots to visualize the coefficient variation of normalized median intensity results of reference methods. Within the violin plot, dots inside the shaded areas indicate the allocation of normalized median intensities. The central thick bar highlights the interquartile range, while the extended thin points denote the maximum (upper) and minimum (lower) values of the data.

2, it is seen that Zheng et al. [7], Hoque et al. [9], Salvi et al. [8], and Vahadane et al. [42] exhibit significantly superior performance in calculating normalized median intensities for hematoxylin and eosin stains as compared to other reference methods. The statistical analysis involved utilizing standard deviation (SD) and coefficient variation (CV) of normalized median intensity results from various methods, as presented in Fig. 1. Boxplots are employed to compare the distributions of various state-of-the-art methods based on the standard deviation of normalized median intensity, considering factors such as the shape of the distribution, central value, and variability. In violin plots, as shown in Fig. 2, we combine boxplots and density plots to visualize both the distribution and probability density of the data. This helps to evaluate the coefficient variation of normalized median intensity results from various methods where the shadow represents normalized median intensity allocation. The interquartile range is highlighted by a thick bar at the center, and the maximum (upper) and minimum (lower) values of the data are represented by extended thin points. Figs. 3 and 4 illustrate the performance of color normalization by presenting a comparison of the normalization results obtained from different state-of-the-art methods working on Dataset 1 and Dataset 3 respectively.

The results in Figs. 1, 2, 3, and 4 demonstrate that Reinhard et al. [20] changes the color space from RGB to $\ell\alpha\beta$, which is appealing, but the assumption of uniform color distribution in each channel leads to poor normalization in the least dominant channel, even though the source intensity variation is preserved. The Macenko et al. [6] uses a color deconvolution approach to find the stain metric, but if the stain metric estimation fails, the method fails as well. The method's

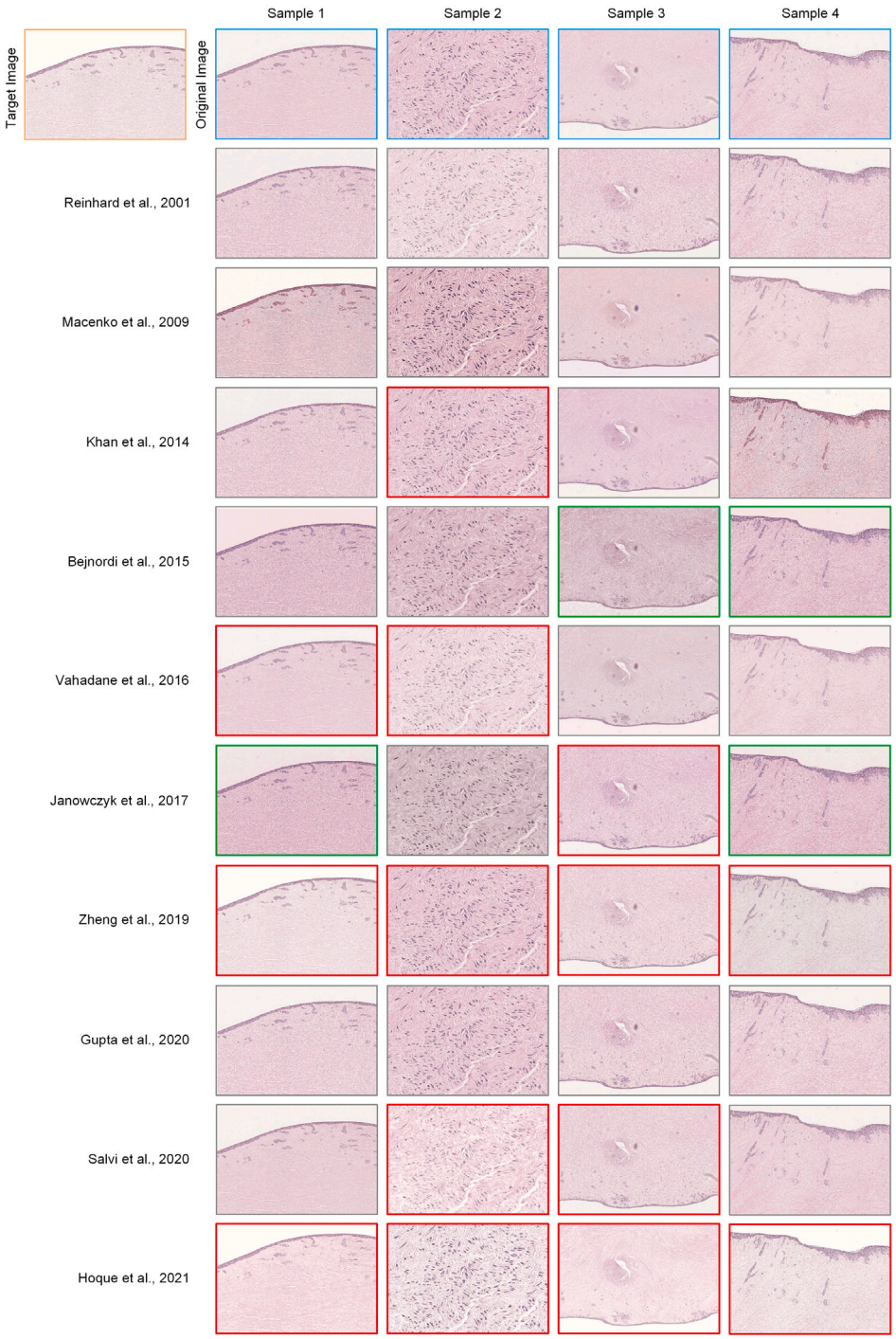


Fig. 3. The normalization results from the compared methods for four myoma sample images from dataset 1 where the green boxes indicate the results that have apparent artifacts and the red boxes indicate results that are free of artifacts or any noticeable color discontinuity.

deviation from the target image increases when the concentration of the stain changes non-linearly. The stain color descriptor approach, Khan et al. [18] uses a supervised linear mapping and classification approach to find the stain matrix, which may not work well if the contrast of the source image is low. This method also cannot preserve all biological information in the source image. Bejnordi et al. [36] removes color bias due to illuminant variation in the histopathology image, but relies on the target image and may not preserve all source intensity variations. The structure preserving color normalization approach, Vahadane et al. [42] transfers the mean color of the target image to the source image while preserving all histological information through non-matrix factorization (NMF). Although the performance results are good the computational complexity is comparatively higher. Additionally, the Beer–Lambert law of absorption is not valid for the complex color appearance [35], therefore this method cannot preserve all color information of source image. Janowczyk et al. [31] produces favorable results, but it encounters some difficulties due to artifacts present in the normalized images, resulting in a lack of smoothness in the results. These artifacts are observed to affect the color distribution of the monochromatic whole slide image which led to an uncertain

Table 4
The performance comparison results of stain separation (mean ±standard deviation), relative square error of hematoxylin and eosin stain estimation, time complexity and average running time of the reference methods.

Datasets	Methods	Stain Separation (Mean ± STD)		Relative Square Error		Time Complexity		Time (s)
		S_H	S_E	rSE_H	rSE_E	*Preprocessing*	Model solution	
Dataset 1	Reinhard [20]	0.931 ± 0.029	0.208 ± 0.033	0.0089	0.0060	–	$\sigma(n)$	3.91
	Macenko [6]	0.915 ± 0.055	0.351 ± 0.050	0.0086	0.0051	–	$\sigma(n)$	14.22
	Khan [18]	0.872 ± 0.082	0.312 ± 0.063	0.0053	0.0039	$\sigma(n)$	$\sigma(n)$	1210
	Bejnordi [36]	0.853 ± 0.098	0.306 ± 0.057	0.0081	0.0168	$\sigma(n^3)$	$\sigma(n)$	489
	Vahadane [42]	0.942 ± 0.025	0.218 ± 0.039	0.0030	0.0045	–	$\sigma(n)$	18.1
	Janowczyk [31]	0.904 ± 0.075	0.279 ± 0.072	0.0064	0.0073	–	$\sigma(n)$	73.16
	Zheng [29]	0.848 ± 0.081	0.381 ± 0.083	0.0092	0.0115	–	$\sigma(nlog_2n)$	42.01
	Zheng [7]	0.960 ± 0.034	0.452 ± 0.037	0.0025	0.0031	–	$\sigma(n)$	2.89
	Gupta [45]	0.911 ± 0.032	0.425 ± 0.061	0.0038	0.0089	–	$\sigma(n)$	4.40
	Salvi [8]	0.951 ± 0.026	0.311 ± 0.041	0.0027	0.0039	–	$\sigma(n)$	3.19
	Hoque [9]	0.976 ± 0.016	0.485 ± 0.014	0.0011	0.0017	–	$\sigma(n)$	2.11
Dataset 2	Reinhard [20]	0.932 ± 0.050	0.317 ± 0.057	0.0044	0.0030	–	$\sigma(n)$	3.55
	Macenko [6]	0.947 ± 0.042	0.349 ± 0.025	0.0019	0.0024	–	$\sigma(n)$	11.10
	Khan [18]	0.940 ± 0.022	0.319 ± 0.033	0.0042	0.0036	$\sigma(n)$	$\sigma(n)$	511.02
	Bejnordi [36]	0.894 ± 0.053	0.297 ± 0.042	0.0074	0.0114	$\sigma(n^3)$	$\sigma(n)$	190.73
	Vahadane [42]	0.942 ± 0.020	0.311 ± 0.029	0.0037	0.0031	–	$\sigma(n)$	7.83
	Janowczyk [31]	0.887 ± 0.059	0.253 ± 0.051	0.0062	0.0088	–	$\sigma(n)$	55.12
	Zheng [29]	0.865 ± 0.033	0.284 ± 0.064	0.0073	0.0090	–	$\sigma(nlog_2n)$	21.36
	Zheng [7]	0.971 ± 0.011	0.492 ± 0.011	0.0010	0.0013	–	$\sigma(n)$	2.02
	Gupta [45]	0.939 ± 0.037	0.391 ± 0.015	0.0035	0.0028	–	$\sigma(n)$	2.39
	Salvi [8]	0.959 ± 0.023	0.455 ± 0.027	0.0018	0.0021	–	$\sigma(n)$	2.15
	Hoque [9]	0.964 ± 0.017	0.473 ± 0.014	0.0015	0.0019	–	$\sigma(n)$	1.79

Table 5
The quality performance comparison in terms of QSSIM, SSIM, and PCC.

Methods	Dataset 1			Dataset 2		
	QSSIM	SSIM	PCC	QSSIM	SSIM	PCC
Reinhard et al. [20]	0.927	0.922	0.930	0.953	0.934	0.941
Macenko et al. [6]	0.939	0.934	0.941	0.950	0.0946	0.934
Khan et al. [18]	0.902	0.847	0.879	0.924	0.929	0.918
Bejnordi et al. [36]	0.823	0.79	0.837	0.819	0.837	0.841
Vahadane et al. [42]	0.946	0.925	0.934	0.944	0.930	0.938
Janowczyk et al. [31]	0.881	0.872	0.896	0.923	0.799	0.872
Zheng et al. [29]	0.873	0.855	0.912	0.923	0.916	0.902
Zheng et al. [7]	0.959	0.952	0.961	0.967	0.956	0.971
Gupta et al. [45]	0.939	0.951	0.954	0.932	0.909	0.947
Salvi et al. [8]	0.963	0.959	0.950	0.952	0.949	0.943
Hoque et al. [9]	0.975	0.982	0.984	0.955	0.948	0.960

rate of failure. However, the methods proposed by Macenko et al. [6], Bejnordi et al. [36], Zheng et al. [29] and Gupta et al. [45] perform well in color normalization, but they may fail due to color artifacts and variance. Janowczyk et al. [31], Khan et al. [18] and Reinhard et al. [20] methods yield good results, but the normalized images often contain various artifacts, resulting in a non-smoothed color appearance. These methods typically face difficulties in dealing with color distribution in monochromatic histopathology images. Normally, when pixels of similar color in the original images are separated into different classes and normalized, it may lead to noticeable apparent color discontinuities in the resulting normalized images. Such color discontinuities particularly appear on the boundary between hematoxylin and eosin stains, as illustrated in Figs. 3 and 4 (highlighted with green boxes). In contrast, unified-transformation-based techniques incorporate prior knowledge of hematoxylin and eosin-staining into their models to estimate a unified transformation for all pixels. These approaches mitigate the issue of color discontinuity and structural artifacts, resulting in more consistent and precise normalized images.

It is seen that Hoque et al. [9], Zheng et al. [7], Salvi et al. [8] and Vahadane et al. [42] outperform all other reference methods by effectively removing color artifacts, estimating hematoxylin, eosin, and background distributions, and classifying pixels with better color distribution in monochromatic whole slide images. These approaches provide the best results among all the methods, as these methods accurately separate stains in both source and target images. A detailed evaluation of staining separation (mean ±standard deviation), relative square error of hematoxylin and eosin, time complexity, and average running time of the reference methods are presented in Table 4.

In terms of standard deviation and relative square error, Zheng et al. [7], Hoque et al. [9], and Salvi et al. [8] consistently outperform the other reference methods. Table 4 also compares the time complexity of each method based on the number of pixels and average running time. It is seen that the pixel-wise classification method, Khan et al. [18] is linearly related to the number of pixels, n which has linear time complexity $\sigma(n)$. Bejnordi et al. [36] has complex cubic time complexity as $\sigma(n^3)$ due to the incorporation of Hough transform. Zheng et al. [29] utilizes a sorting algorithm for n value to estimate parameters, which results in a time complexity of $\sigma(n\log_2 n)$. Reinhard et al. [20], Macenko et al. [6], Vahadane et al. [42], Janowczyk et al. [31] and Gupta et al. [45] methods with time complexity $\sigma(n)$ estimate the parameters by using pixel-wise operation. In this case, Hoque et al. [9], Zheng et al. [7] and Salvi et al. [8] have relatively short average normalization parameter estimation running times on both Dataset 1 and 2. In our experiment, QSSIM, SSIM, and PCC are used to evaluate and compare the quality performance of the reference methods on various myoma and lung samples.

Table 5 shows that Zheng et al. [7], Hoque et al. [9], and Salvi et al. [8] exhibit comparatively higher QSSIM, SSIM, and PCC scores on both datasets compared to other reference methods. In summary, the experimental results obtained from Dataset 1 and 2 (Tables 1–4, and Figs. 1–3) indicate that the methods proposed by Hoque et al. [9], Zheng et al. [7], Salvi et al. [8], and Vahadane et al. [42] exhibit superior performance in stain normalization, with fewer apparent artifacts in normalized images. Moreover, the experimental findings from Dataset 3 (Table 6 and Fig. 4) suggest that the methods developed by Salvi et al. [8], Hoque et al. [9], and Zheng et al. [7] consistently outperform other reference methods. From our experimental comparison, it is clear that the unified-transformation-based techniques that preserve the structure of stain color appearance are significantly more effective than other current state-of-the-art methods.

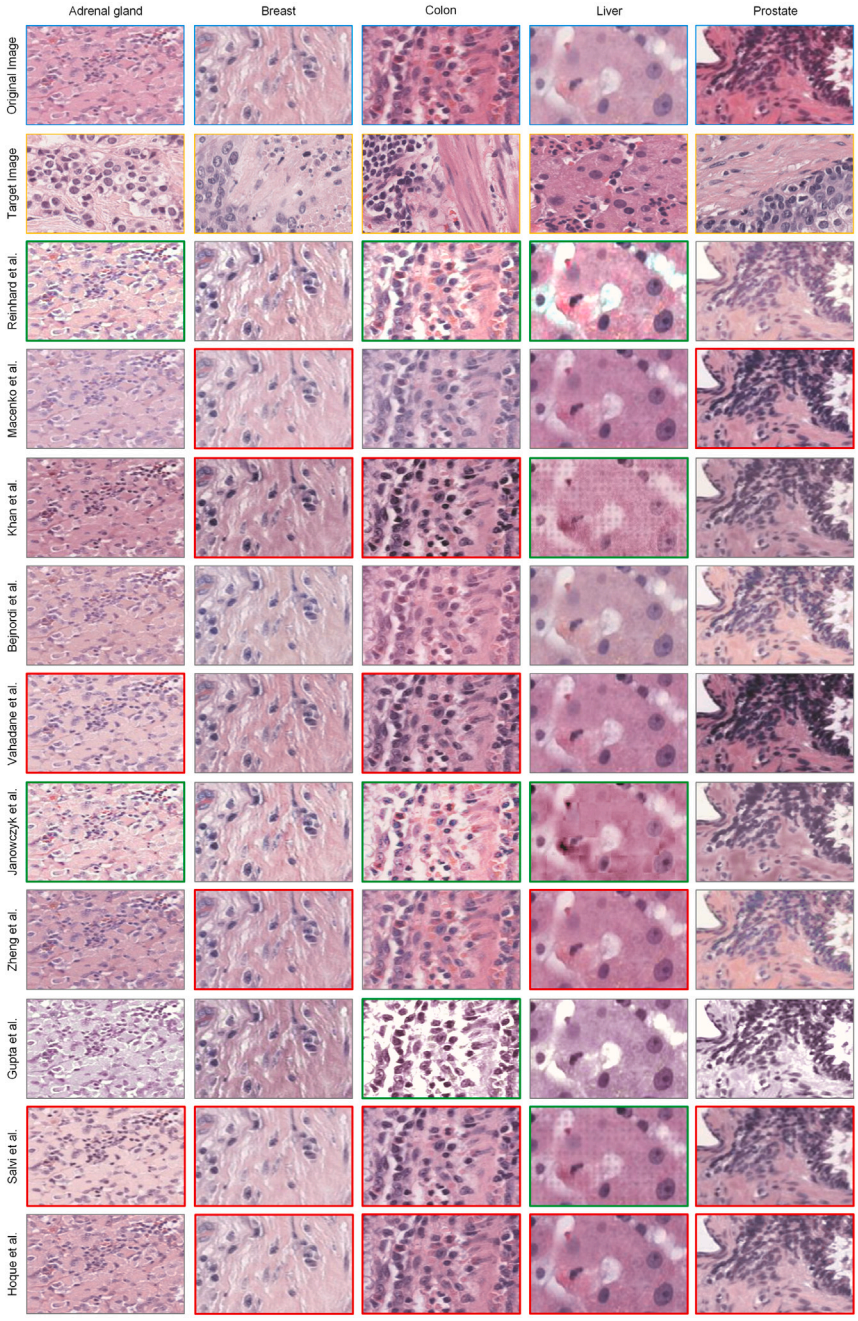


Fig. 4. The normalization results from the compared methods for five sample images from dataset 3 where the green boxes indicate the results that have apparent artifacts and the red boxes indicate results that are free of artifacts or any noticeable color discontinuity.

Table 6
The performance comparison results of relative square error of hematoxylin and eosin stain estimation working on different tissue types from dataset 3 using various methods.

Methods	Adrenal		Breast		Colon		Liver		Prostate	
	rSE_H	rSE_E	rSE_H	rSE_E	rSE_H	rSE_E	rSE_H	rSE_E	rSE_H	rSE_E
Reinhard et al. [20]	0.0181	0.0073	0.0044	0.0110	0.0091	0.065	0.0062	0.0050	0.0029	0.0047
Macenko et al. [6]	0.0119	0.0086	0.0086	0.0172	0.0094	0.0071	0.0088	0.0053	0.0061	0.0097
Khan et al. [18]	0.0015	0.0042	0.0009	0.0154	0.0022	0.0022	0.0030	0.0005	0.0051	0.0032
Bejnordi et al. [36]	0.0071	0.0105	0.0043	0.0015	0.0029	0.0024	0.0031	0.0068	0.0018	0.0022
Vahadane et al. [42]	0.0031	0.0027	0.0013	0.0036	0.0014	0.0023	0.0024	0.0045	0.0012	0.0017
Janowczyk et al. [31]	0.0051	0.0067	0.0055	0.0039	0.0020	0.0017	0.0031	0.0025	0.0013	0.0018
Zheng et al. [29]	0.0095	0.0112	0. 0041	0.0120	0.0061	0.0029	0.0023	0.0049	0.0021	0.0027
Zheng et al. [7]	0.0007	0.0009	0.0010	0.0014	0.0008	0.0010	0.0007	0.0014	0.0008	0.0005
Gupta et al. [45]	0.0049	0.0033	0.0027	0.0020	0.0021	0.0035	0.0019	0.0015	0.0013	0.0010
Salvi et al. [8]	0.0008	0.0009	0.0005	0.0025	0.0009	0.0011	0.0005	0.0010	0.0010	0.0007
Hoque et al. [9]	0.0010	0.0009	0.0006	0.0011	0.0007	0.0010	0.0005	0.0008	0.0010	0.0006

6. Discussion and future directions

The interpretation of histological specimen images by pathologists is partially limited by color variability. Such variability can significantly impact the performance of automated image analysis algorithms. In this regard, stain normalization techniques are widely used in digital pathology to standardize the color appearance of tissue samples. Our study summarizes various color normalization methods and highlights the most effective techniques with an experimental comparison. Taking into consideration the publication summaries given in Table A.1 and the experimental comparison in Section 4, it is clear that various stain normalization techniques carry unique strengths and limitations. Our experimental observation indicates that these techniques have been designed to address specific challenges associated with staining. The experimental findings of our study indicate that color normalization methods that preserve image structures, including those proposed by Hoque et al. [9], Zheng et al. [7], Salvi et al. [8], and Vahadane et al. [42] yielded superior outcomes. However, it is important to note that this does not imply the ineffectiveness or substandard nature of other reference methods. The efficacy of each technique is dependent on the specific staining difficulty it is intended to address, and each method has its respective advantages and limitations. Furthermore, our thorough investigative study found that state-of-the-art methods performed exceptionally well on their respective datasets for which they were designed. The most important thing is that stain normalization techniques have the potential to enhance the accuracy and consistency in research projects and even in diagnostic digital pathology by improving the reproducibility and comparability of images.

The significance of stain normalization techniques is especially pronounced in critical diagnostic decisions as well as large research projects combining stained slides from several locations as they reduce variability arising from differences in staining protocols and imaging conditions. Different stain normalization techniques improve efficiency by reducing the need for manual adjustments of color balance or intensity, which is particularly beneficial for large scale whole slide image analysis. Moreover, these techniques facilitate tracking of disease progression or response to treatment by standardizing the analysis process, making the images directly comparable. They also improve the visualization of digital images, making it easier to identify relevant features or anomalies. Additionally, they enable quality control, reduce inter-observer variability, and allow for automated analysis, thereby enhancing the efficiency and effectiveness of diagnosis, research, and patient care. In recent times, several normalization methods have been proven highly effective in addressing detection (e.g., cells, nuclei, or epithelial regions), segmentation, and classification challenges [65–70]. As with any technological innovation, there are potential drawbacks to the use of stain normalization. One major challenge is the possibility of losing important information during the normalization process, which can affect the accuracy and reliability of results. Furthermore, unintended consequences of image normalization may occur, including the masking of vital biological features that are critical for diagnosis or research. In some cases, normalization may not be suitable for certain staining protocols, leading to suboptimal results or introducing bias into the analysis.

Over-normalization is another issue that can arise when an aggressive algorithm attempts to remove too much variability, resulting in artificially homogeneous images that are less representative of the original sample. This may cause the loss of significant diagnostic information. The lack of standardization for stain normalization in digital pathology can also make it challenging to compare results across studies or develop standardized protocols. Normalization may also be limited in its ability to correct staining artifacts, such as uneven or non-specific staining. This is due to the assumption that staining is consistent across the image, which may not always be the case. As a result, staining artifacts may still be present in the normalized image, which can impact the accuracy of downstream analysis. Optimizing different tissues may also cause challenges, as they may have varying staining characteristics that may require different normalization techniques. Moreover, validating normalized results can be difficult, particularly in the absence of ground truth for comparison, as normalization typically involves subjective decisions such as selecting regions of interest or adjusting normalization parameters. Consequently, it can be challenging to determine the accuracy or reliability of the normalization process.

In conclusion, while stain normalization provides numerous benefits and potential drawbacks, it is essential to carefully consider the risks and limitations of the technique when applying it to histopathology data. The selection of the appropriate normalization technique should be based on the specific application and the characteristics of the images being analyzed. Each stain normalization technique has its unique advantages and limitations, and its effectiveness depends on factors such as the staining protocol, imaging conditions, types of data, specific research questions, and requirements of the study. Because of this, it is imperative that digital pathology experts thoroughly evaluate the impact of any normalization or feature extraction technique on the downstream analysis, and choose the approach that is most appropriate for the specific application. To determine the most suitable methodology for analyzing histopathology data, it is recommended to try different techniques and carefully consider which one best fits the research question and data at hand. Thus, it is important to conduct a comprehensive evaluation of the technique's suitability for a specific application to ensure accurate and reliable results. Although several stain normalization techniques have been proposed in the literature, there is still room for further research [1–4,71,72]. One direction for future research can be directed toward the exploration of deep learning techniques, which can learn the mapping between different color spaces and enable efficient color normalization. Another potential direction is to investigate the use of unsupervised learning techniques, such as clustering or dimensionality reduction, to identify common color patterns across different tissue types and use this information to guide the normalization process.

Furthermore, as digital pathology is becoming more widespread and integrated into the clinical workflows, there is a growing need for fast and efficient stain normalization methods that can be applied to large-scale datasets. Consequently, future research can be focused on developing computationally efficient algorithms that can perform stain normalization in real-time or on distributed computing architectures. In addition, the use of multispectral imaging technology allows the capture of images at multiple wavelengths, which can be used to distinguish between different tissue components and enhance the contrast between stains. By leveraging the additional information provided by multispectral imaging, it may be possible to develop more accurate and robust stain normalization methods. In addition, there is a need for stain normalization techniques that are robust to variations in tissue morphology and staining quality. One way to achieve this could be incorporating information about tissue structure and staining quality into the normalization process, e.g., using morphological features or stain intensity distributions. By taking into account these additional factors, it is possible to develop new methods that are more effective in a wider range of settings.

Finally, as the field of digital pathology continues to develop, there may be new challenges and opportunities that arise for stain normalization. For instance, as more data becomes available from different sources and with different staining protocols, there may be a need for methods that can adapt to these variations in a more flexible and automated way. Therefore, future research may also need to focus on developing methods that are more adaptable and scalable to different settings and applications. Most importantly, the choice of the method depends on the specific requirements of the application, including image complexity, training data availability, and computational resources. Typically, stain normalization techniques can indeed introduce certain levels of noise or artifacts in an image, and it is essential to consider the potential downstream effects when selecting a technique. It is worth

noting that alternative approaches such as color augmentation and domain adaptation may also introduce noise or artifacts in the images, depending on the chosen specific approach.

One alternative approach that can be effective in some cases is to use stain-invariant features. These features are derived from the image and designed to be insensitive to variations in staining. The features can be extracted and then used directly for downstream analysis or as input to machine learning models. This approach has been shown to be effective in some applications, particularly in situations where the staining variability is moderate and the aim is to identify patterns in the tissue rather than the specific stain. Another approach is to use unsupervised or self-supervised learning techniques that do not require explicit normalization of the images. For example, deep clustering and contrastive learning techniques can learn image representations that are insensitive to variations in staining, by encouraging similar images to be mapped to nearby points in the feature space. This approach has shown promising results in some applications, particularly in situations where the variability in the staining is high, and traditional normalization techniques may not be effective.

Our overall observation is that, while existing methods are not sufficiently accurate or robust for routine automatic clinical trials, they highlight a starting point for further research aimed at finding better solutions. Therefore, addressing the complexities of stain normalization in complex histopathology images requires further research to refine existing techniques and develop novel approaches.

7. Conclusion

In digital pathology, the use of whole slide imaging and artificial intelligence has improved computer-aided diagnosis. But the inconsistent staining color due to differences in staining methods and scanning equipment can significantly affect the accuracy of computer-aided diagnosis and can influence the diagnosis made by pathologists. To address this challenge, an effective stain normalization method can standardize color appearances and reduce variability in histopathology images. There are several stain normalization techniques that can notably improve the accuracy, consistency, and efficiency of histopathology image analysis. By minimizing variability and enhancing image comparability, these techniques can significantly enhance the effectiveness of diagnostic processes, research endeavors, and patient care. These techniques are developed to solve specific challenges that often arise during the staining or scanning process and each of these techniques comes with it's strengths and limitations.

This paper presents a comprehensive review and experimental comparison of different stain color normalization methods. Our experimental findings reveal that stain normalization tends to introduce significant artifacts in the output images, rendering it inapplicable when the statistics of the target and source images are dissimilar. The mean brightness of the original image is not maintained by several stain normalization methods, even though they preserve the source intensity variation. Some methods are unable to produce results that are consistent and closely match the target image. These methods may cause significant variations compared to the expected normalization results. Additionally, in low-contrast images, several methods fail to preserve the structure and intensity variation of the source image. Our findings indicate that the structure-preserving unified-transformation-based techniques including those proposed by Zheng et al. 2019, Hoque et al. 2021, Salvi et al. 2020, and Vadhane et al. 2016 are superior compared to different state-of-the-art methods. These methods preserve both the brightness and structure of the source image and produce fewer artifacts than existing state-of-the-art methods.

However, our observation is that each of the existing stain normalization techniques still has some potential drawbacks, such as the loss of critical information during the normalization process and over-normalization, which can lead to the masking of important biological features. Therefore, the selection of the appropriate stain normalization technique must be based on careful consideration of the specific application and the characteristics of the images being analyzed. In this context, digital pathology experts must thoroughly evaluate the impact of any normalization or feature extraction technique on the downstream analysis and choose the approach that is most appropriate for the specific application. Further research is needed to optimize stain normalization techniques and evaluate their influences on the steps of the computer-aided diagnostic decision support system (mainly comprised of segmentation, feature extraction, and classification steps). A thorough comparison of existing methods, including their working principles, advantages, limitations, and possible future research indicated by this paper will encourage the investigation of new methods. The results of this comprehensive review emphasize the importance of further research in this domain, with a specific focus on the development of innovative techniques that can improve the accuracy of computer-aided diagnosis systems and have potential applicability in medical practices.

Ethical approval

The Ethical Committee of Northern Ostrobothinia Hospital District and the Scientific Committee of Biobank of Eastern Finland approved the research (the use of myoma and lung tissue samples).

CRediT authorship contribution statement

Md. Ziaul Hoque: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Writing – original draft. **Anja Keskinarkaus:** Investigation, Data curation, Writing – review. **Pia Nyberg:** Resources, Image selection, Writing – review. **Tapio Seppänen:** Supervision, Project administration, Resources, Funding acquisition, Writing – review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A.1 represents a summary of the working principles, advantages, and disadvantages of different stain normalization methods used in histopathology image analysis.

Table A.1
Working principles, advantages and disadvantages of different stain normalization methods used in whole slide image analysis.

Methods	Descriptions	Advantages	Limitations
Reinhard et al. [20]	• Color transformation based method. • A statistical analysis to impose color characteristics of one image to another. • Converts the RGB image to $\imath\alpha\beta$ space for transferring background color from target to source image. • Finally, converts the image from $\imath\alpha\beta$ space to RGB space to achieve color correction.	• Helps to manipulate RGB images. • Achieves color correction by choosing an appropriate source image and applying its characteristic to another image. • Preserves all the structure of the source Image. • The contrast of the processed image is approximately as similar as the contrast of the target image.	• Difficult to separate the stains properly due to the transformation in $\imath\alpha\beta$ space. • Describes hue correction using gray world assumption not the Gamma correction because the mean and standard deviations are manipulated in $\imath\alpha\beta$ space as followed by transforming back to RGB space.
Ruifrok et al. [21]	• Color deconvolution method. • Converts RGB image to optical density space. • The stain color appearance matrix is empirically obtained by estimating the relative color ratio for R, G, and B channels through a single stain. • The stain depth matrix is estimated based on the inverse of stain color appearance matrix and multiplied with the value of intensities into the optical density space.	• First attempt to separate stains in histopathological images. • Color ratios of hematoxylin and eosin stains can easily be estimated. • The histochemical procedures never influence the amount of stain in the sample due to stain solubility and prevention of staining saturation.	• A supervised technique with higher computation complexity. • Prior information (single stained slide) is needed to estimate the color appearance matrix. • There is no way to get back the original color of staining, once the color histogram structure is ruined by non-linear mixing of hematoxylin and eosin stains.
Rabinovich et al. [73]	• Spectral decomposition by non-negative matrix factorization and independent component analysis approach. • Factorization of stained images into stain depth matrix and color appearance matrix where the coefficient of color appearance matrix and stain depth matrix are positive. • A joint approximate diagonalization of Eigen metrices to recover independent components for the decomposition of independent component analysis.	• Unsupervised technique which cannot estimate the coefficients of color appearance matrix separately. • There is not any physical interpretation of negative coefficient.	• There is scaling ambiguity so that the solution of non-negative matrix factorization is not closed to the expectation. • Non-negative matrix factorization does not carry any information about image formation. • It is assumed that each stain acts independently into the image for the independent component analysis which is a wrong statement.
Macenko et al. [6]	• Fringing technique that converts the RGB image to the optical density space. • Creates plane corresponding to the largest singular values and find the corresponding angles. • Maximum and minimum angles are estimated by the percentile, α^{th}, and $(100-\alpha)^{\text{th}}$ to find out the robust estimation. • Finally, these extreme values are transferred back to the optical density space.	• It does not have the ambiguity in its solution due to the non-negative matrix factorization. • There is not any negative coefficient in the color appearance matrix.	• Not applicable for automatic cancer detection because the extreme angle values are estimated empirically. • Using this algorithm, it is hard to preserve all the information of source image resulting in a loss of source information in the processed image.
Gavrilovic et al. [74]	• Blind color decomposition approach by separating intensity information from color through blind color decomposition. • Converts RGB image to Maxwellian color space to find the color distribution of the separated stains. • Linear decomposition to estimate stain absorption vectors by identifying the reference color vectors and adjust color variation based on those vectors.	• An unsupervised approach that does not require any prior information for the color decomposition. • Having better performance on non-negative matrix factorization and stain separation.	• The source intensity information is not preserved by this method which introduces some artifacts in normalized image. • Inaccurate performance when spectral variations in stains occur between source image and target image.

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Table A.1 (*continued*).

Methods	Descriptions	Advantages	Limitations
Khan et al. [18]	• An approach based on nonlinear mapping that drives principle color histograms from the quantized histograms. • The use of color based classifier that incorporates the global stain color deconvolution specific color descriptor to calculate image specific stain matrix. • Generates stain probability maps and finds color of each stain by learning a supervised classification framework. • A regularized spline based function is estimated to map the statistics of source image channel to the statistics of corresponding channel in target image.	• A supervised method that works at pixel level. • Largely independent for each stain. • Produces more appropriate deconvolution matrices than simply using standard matrices which helps to separate the stain more effectively.	• Estimating non-linear spline function in color normalization ruins the structure of original image which produces undesirable artifacts in normalized image. • There is no way to get back the original image structure, once the color structure is ruined by non-linear mixing of the stains. • The computational complexity is comparatively higher as for the supervised method.
Magee et al. [19]	• Two approaches including a multi-modal extension to linear normalization in $\iota\alpha\beta$ color space and normalization in a representation space obtained by stain-specific color deconvolution. • A probabilistic prior and a Variational Bayesian Gaussian Mixture Model are used to achieve per-pixel stain for a unique transformation. • A multi-modal color deconvolution based approach is introduced to solve the problems of linear normalization in $\iota\alpha\beta$ color space.	• Preserves the color variation and structural information present in the original image. • Able to solve the problems of linear normalization in $\iota\alpha\beta$ color space.	• Poor consistency with the unimodal prior. • This approach is not accurate enough where color information may act as noise or artifacts after normalization. • The presence of strong color variations may cause failure.
Mccann et al. [75]	• A wedge finding approach that finds a wedge of pixel values for stained images (maximum and minimum values are either hematoxylin or eosin). • Separates stain as similar to Macenko et al. [6] only difference is that, in this approach, eosin is assumed as lower contrast (minimum value) and hematoxylin is assumed as higher contrast (maximum value).	• Assuming maximum and minimum values is a good way to reduce solution space of general fringing method. • This approach does not have any negative coefficient in stain color appearance matrix.	• This method is not suitable for computer-aided diagnosis due to the evaluation of extreme angle values empirically. • It is difficult to preserve all the source color information in the processed image.
Bautista et al. [76]	• Color standardization by color correction matrix approach where image is converted from non-linear to linear RGB. • Color correction is completed in linear RGB space with gamma correction where color correction matrix is calculated by the least square error between target and source images. • Finally, the processed image gets back to the original RGB space.	• Easy to compute due to the global transformation, not the pixel to pixel transformation. • The color and brightness statistics of processed images are comparatively better.	• The differences between target and source images are estimated using least square error through which all the color variations of source image are replaced with target image. As a result, several color artifacts may possibly appear in the processed image. • It is hard to separate stains instead of performing transformation in a linear RGB color space.
Li et al. [28]	• An illuminance matching technique to adjust color variation (both illuminant and stain variations) in stained histo-pathology images. • Identifies the causes of color variation by using illuminant and spectral normalization modules. • A non-negative matrix factorization based stain decomposition is developed to decompose the stains. • Saturation-weighted statistical method is employed to reduce the solution space of non-negative matrix factorization and color normalization is performed where the stain depth matrix is preserved.	• Reduces the solution space of non-negative matrix factorization by using Saturation-weighted statistics. • Converts the image to color appearance matrix or saturated image which is advantageous to limit the impacts of achromatic pixels on color estimation. • Produces consistent stain estimation with preserving capability of histological information (e.g. texture details, spatial structures etc.). • The color histogram structure of the original image is preserved and never ruined by producing undesirable artifacts.	• Saturation-weighted statistics depend on dominant color which is not a natural process. • Contrast has never been enhanced for the fed images. • Inappropriate to address color variation caused by non-standard imaging conditions. • It only preserves the brightness intensity not all of the color information.
Tam et al. [77]	• A centroid alignment based method that normalizes intensity range of image with the help of centroid alignment. • The image is divided into blocks and each block intensity histogram is mapped to the target histogram.	• Illuminance variation is corrected by performing a centroid shift in the intensity distributions. • Removes the batch effects and preserves the modality of the intensity distributions.	• Depends on spatial dependency of pixels. • Local histogram specification does not work effectively and unable to preserve all source information into normalized image.

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Methods	Descriptions	Advantages	Limitations
Vahadane et al. [42]	• A sparse stain separation and structure preserving approach where sparseness is added to reduce solution space of non-negative matrix factorization. • A block coordinate descent algorithm is used to solve the non-convex optimization problem. • A structure preserving scheme is employed to keep source structural information intact.	• Solution space of non-negative matrix factorization can be reduced. • Only changes the color of one image to another and preserves all the structures of original image. • Estimates global color appearance and outperforms all the previous supervised stain separation approaches.	• The computational complexity is comparatively higher though the solution space of non-negative matrix factorization is reduced by this technique. • The Beer–Lambert law of absorption is not valid for the complex color appearance, therefore this method cannot preserve all color information of source image. • There is high possibility of solution optimization problems that might reach to the local minima rather than global minima.
Bejnordi et al. [36]	• A hue saturation density based approach to standardize image color. • Transforms RGB to the chromatic and density components. • The chromatic and density distribution for each component (hematoxylin, eosin and background) in hue saturation density are aligned to match the corresponding distributions. • After weighting the stain for each pixel, it transforms the hue saturation density color to RGB color.	• Detects the stain components by using an unsupervised approach and preserves spatial information. • Preserves the structures of original image.	• High processing time with computational complexity. • All the information of source image is not preserved in processed image (e.g. red spots). • After normalization the background color of original image is removed. • Comparatively good in normal stain variations but bad in strong stain variations.
Alsubaie et al. [10]	• Independent component analysis and wavelet decomposition based approach. • The images are decomposed into wavelet sub bands and source stains are considered as independent components. • The independent components are separated with the help of independent component analysis. • Statistically analyzes the multi-resolutional representation to separate independent observations and estimate the stain matrix.	• An unsupervised technique. • Stain color consistency is not applicable after color deconvolution. • Sources are more independent and separable by filtering the original optical density image. • The method is mainly concerned with stain color deconvolution more than color estimation.	• The wavelet transformation reduces redundancy in image pixels. • It is hard to make sure the source components exactly independent. • Independent component analysis is applicable only when the stains are completely independent of each other.
Zarella et al. [78]	• Stain color normalization approach based on cluster's centroid. • Converts RGB image to HSV color space where each pixel represents hue, saturation and value. • Performs clustering by the transformation of pixel values into a Cartesian coordinate system. • Maps each pixel in a cluster to the HSV value of the cluster's Centroid to establish a ten-color image.	• Preserves all the structures of original image. • Color variability is reduced in hue, saturation and value by a factor of 6-16, and white space variability is reduced by a lesser factor (approximately 2) for saturation and value.	• Dominant colors (e.g. blue and pink) are transferred or preserved from the target image to source image. • There is a high possibility to lose important information due to the dominant color transformation. • Depends on two dominant colors including blue and pink only and hue is not a meaningful quantity for white space.
Janowczyk et al. [31]	• A stain normalization approach using Sparse Autoencoders. • Training of unsupervised deep Sparse Autoencoders to partition images into sub-types and standardize color independently. • A learned filter and an unsupervised clustering approach are applied on source images to identify the centers of K cluster. • Subsequent histogram matching and alignment across clusters with different color channels.	• Reduces inter scanner color variance. • Reduces extreme color variations induced by slide preparation and digitization, differential staining etc. • The overall performance is comparatively better with the combination of global normalization and Sparse Autoencoders techniques.	• There is a high chance to lose some important information due to multiple image processes (filtering and unsupervised clustering). • It might cause wrong normalization results due to the reduction of mean and standard deviation by using global normalization and Sparse Autoencoders techniques.

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Methods	Descriptions	Advantages	Limitations
Bentaieb et al. [46]	• A discriminative model equipped with a stain normalization component that transfers stains across datasets. • Comprises a generative network to learn dataset specific staining properties and image specific color transforms as well as task specific discriminative models. • Stain normalization components are included to perform non-linear mapping between two images in different distributions. • Stain transformation is performed with encoder and decoder architectures.	• A fully trainable joint normalization and leveraging adversarial model. • The low-level information related to texture present in the original image can be preserved effectively. • The structural information is also preserved with the help of regularization loss.	• The method does not rely on any pre-processing step, it only involves training deep networks which creates complexity to estimate dataset distributions accurately. • Fails to provide the desired color normalization outcome, and the processed image differs from the target image. • There is a high chance to lose some important information during color normalization process.
Zanjani et al. [49]	• A parametric and fully unsupervised generative model based on end-to-end machine learning in the framework of generative adversarial networks. • Converts RGB image into CIEL*a*b color coordinates. • A generator network is used to generate colorized hematoxylin and eosin image. • An auxiliary network is used to estimate the element of transformed hematoxylin and eosin image noise at its output. • The discriminator network distinguishes generated color image from original image. • Finally, get back to the CIEL*a*b color space to RGB color space.	• Preserves the structures of the source image having high mutual chromatic information with a template image. • The source intensity information and the brightness of source image are preserved properly. • Increases the contrast of the final output normalized image.	• It is difficult to separate the stains in the CIEL*a*b color space properly. • It only relies on the color of target image, not the mean color of target image which creates undesirable artifacts in normalized image.
Shaban et al. [51]	• A deep learning approach inspired by Cycle Adversarial Networks. • An unpaired Image-to-Image Translation using Cycle-GANs. • Style-transformation is applied to solve the classic stain normalization problems.	• There is a high visual similarity appearance to the target domain. • This end-to-end learning picks the reference template automatically and avoids the need for picking manually.	• The performance needs to be improved and an unified representation might be the best solution for the improvement. • This approach is not suitable for the many to many style domains of the stained images.
Zheng et al. [29]	• An automatic content-based histopathological image retrieval approach. • Input images are segmented into irregular sub-regions using a linear spectral clustering (LSC) technique [79]. • Histogram transformation based approach. • Color deconvolution method [21] is used to separate the hematoxylin and eosin components. • The luminance and saturation are normalized into the HSV (hue-saturation-value) space prior to color deconvolution method.	• The hematoxylin and eosin stain color ratios can easily be estimated. • The global illumination and saturation of staining are normalized. • Eliminates the variances of illumination and saturation.	• Single stained slide is needed to estimate the color appearance matrix. • There is no way to get back the original color of the staining, once the normalization process is done. • Unable to effectively transform the color and various artifacts appear in the processed image. • The normalization performance is limited.
Zheng et al. [7]	• Adaptive color deconvolution algorithm for stain and color separation from stained whole slide images. • A group of pixels is sampled from the tissue region and converted into optical density space. • Normalized hematoxylin and eosin components are obtained with the help of an adaptive color deconvolution matrix and a stain weighted matrix. • Hematoxylin and eosin components are recombined with the specific color deconvolution matrix of a template image. • To avoid artifacts multiple prior knowledge of staining is embedded in the adaptive color deconvolution model. • An integrated optimization is designed that simultaneously estimates the parameters of stain separation and color normalization.	• This method has proven to be effective for color normalization in various color appearances. • Can be applied to whole slide images containing different lesions. • Only involves pixel wise operation that makes it very efficient and applicable to whole slide images. • This method can potentially be applied to stains that satisfy Beer-Lamber law.	• This method does not preserve all the source information with high computational complexity. • Some prior or previous information is required for evaluating the color appearance matrix. • Having limitations in immunohistochemical stains that do not satisfy the Beer-Lamber law. • Difficult to analyze color distribution of the monochromatic whole slide image using this method.

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Methods	Descriptions	Advantages	Limitations
Tellez et al. [80]	• An unsupervised method to perform stain color normalization using a neural network. • Combination of color augmentation and color normalization. • The problem of stain color normalization as an unsupervised image-to-image translation task is formulated to be solved. • An extended morphology with random brightness and contrast image perturbations. • An extended morphology with random brightness and contrast image perturbations and augmentation by randomly shifting hue and saturation channels.	• This approach reduces the generalization error. • The classification performance might be improved drastically by increasing the ability of CNN to generalize the unseen stain variations. • CNNs are mostly insensitive to the intensity of color augmentation. • Preserves computational resources with low computational cost.	• The experiments with grayscale images achieved mediocre performance. • Achieved worst performance with morphology and identity configurations. • Color information may act as noise or artifact without augmentation which increases the overfitting and generalization error due to stain variation.
Tosta et al. [81]	• Image normalization with different degrees of fading using a combination of fuzzy theory and Cuckoo search algorithm. • An estimation that employs fuzzy theory to associate intensities of the pixels with hematoxylin and eosin dyes. • The parameter estimation is carried out using Cuckoo search algorithm. • Global optimum result initialization by applying a technique reported in [28]. • Stain representation using a dictionary learning [82] with an initialization method for optimization.	• Able to preserve the color of the artifact with less vibrant shades of pink than in the original image. • The structures of the original image remain same in the normalized image. • Able to improve the density distribution over the entire interval of [0–255].	• Unable to preserve tissue color information with contrast correction. • Suffers from high computational cost. • Longer estimation time is required.
Roy et al. [32]	• A fuzzy based modified Reinhard color normalization approach to overcome the limitation of conventional Reinhard method [20]. • A global method that satisfies all three hypotheses (correlation coefficient, global mean color and image contrast) of color normalization. • The fuzzy logic is employed to control the contrast enhancement in ι space and color coefficients in $\alpha\beta$ space, in order to reduce color variation. • Estimates the number of pixels associated with homogeneous white luminance portion.	• The effects of color fading are alleviated at a certain level. • All three hypotheses of color normalization are satisfied. • The contrast of the normalized image is approximately as similar as the contrast of the target image. • Less computational complexity.	• Difficult to get back to real numbers space from fuzzy space. • Difficult to separate the stains properly due to the transformation in fuzzy space. • Global estimations are correlated with local metric estimation but there is no guarantee that those global estimations will be same as local estimations.
Gupta et al. [45]	• An approach by utilizing the geometry of color vector space. • Geometry-inspired, chemical-invariant and tissue invariant normalization. • Corrects the illumination variation, stain chemical and stain quantity variation in a unified framework by exploiting the underlying color vector space's geometry. • Benchmarking is done on datasets containing three different cell types and prepared with two different staining chemicals.	• Robust to tissue/cell type and staining chemical. • The efficacy is demonstrated diagnostically in the application of breast cancer detection via a CNN-based classifier.	• The alignment of different origins is required for the illumination corrections. • There are reconstruction errors where the quadratic objective function converges to local minimum instead of global minimum. • Suffers from high computational cost.
Lei et al. [83]	• StainCNNs, an efficient stain feature learning method. • A novel StainCNNs method to accelerate the stain feature evaluation in the SPCN method [42]. • Simplifies the process of stain feature extraction and imply a GPU-enabled realization to accelerate the learning of stain features in the Tensorflow framework. • Deep learning methods to simplify the evaluation of stain color feature matrix, and build the StainCNNs model in the Tensorflow framework.	• Able to make use of the stain feature distribution in dataset. • Able to perform the stain normalization quickly in dataset level. • Combining the advantages of deep learning methods. • Suffers from low computational cost.	• Though StainCNNs technique successfully learns stain features at the dataset level, it lacks an effective stain feature extraction mechanism. • Able to use the stain feature distribution in the dataset but it creates complexity with color artifacts.

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Methods	Descriptions	Advantages	Limitations
Salvi et al. [8]	• Stain color adaptive normalization ($SCAN$) algorithm. • A clustering and segmentation method to utilize the structure detection by preserving dominant colors. • The method is reproducible (it is not based on different initialization metrics). • A refinement of the singular value decomposition based approach. • Pixel wise transformation.	• Able to improve the contrast between tissue and background. • Preserves local structures without changing color of tissue and the background. • Satisfies Beer–Lambert law [35].	• The preliminary stain separation could fail if the dimensions are very small or if one of two stains is not present in the image. • If a stain binds to a few pixels, the preliminary stain separation is not able to detect the specific stain.
Perez et al. [84]	• A method for blind color deconvolution of histo-pathology images based on the total variation. • An extension of previous work proposed in [85] by applying the Variational Bayes inference [86]. • Employs the similarity to a given reference color-vector matrix along with the total variation prior for each stain. • The variational Bayesian algorithm is used to compute color matrix and stain concentration. • It considers spatial relations between pixels by means of a total variation (TV) prior model and the similarity to a reference color-vector matrix.	• The non-quadratic total variation (TV) energy helps to reduce the noise in images while preserving the sharp edges. • The image contrast is increased which meliorates the discrimination power of the eosin band. • Helps to extract relevant information and leads to better classification performance.	• The presence of strong color variations may cause errors. • Making eosin band slightly more contrasted creates undesirable complexity with color artifacts.
Bukenya et al. [87]	• A hybrid approach based on an octree color quantization algorithm combined with Beer–Lambert law, modified blind source separation and color transformation. • This approach includes two stages (stain separation and color transformation). • An octree color quantization algorithm is used to estimate stains based on the chromatic and luminous response. • A modified color transformation algorithm is used to minimize the effect of stain variation and illumination.	• Capable to address color variation problem in both (Haemotoxylin and Diaminobenzidine) and (Haemotoxylin and Eosin) stain images. • Introduces a weighting factor that helps to adjust the contrast and brightness of the normalized image. • This method satisfies the Beer–Lambert law [35].	• Structures and texture of original image might be distorted after normalization. • The normalized image might be deviated from the target image after the contrast adjustment. • High computational complexity.
Nadeem et al. [88]	• An approach to normalize and augment hematoxylin and eosin stained images. • A multimarginal Wasserstein barycenter is used to interpolate the color distributions of source and target images in Lab color space. • Multimarginal Wasserstein barycenter drives both data augmentation and stain normalization.	• The multimarginal setting gives physically more realistic interpolations (augmentation) as well as normalization. • It is independent of the number of pixels so, the method can easily be scaled to histological images.	• Color information may act as noise or artifact without augmentation which introduces overfitting problems. • Longer estimation time is required.
Vijh et al. [33]	• A method that involves fuzzy logic for stain, enhanced fuzzy illuminant and modified spectral normalization. • An enhanced fuzzy modified Reinhard method is introduced for illuminant normalization. • A fuzzy c-mean saturated weight hue statistics is employed for stain normalization. • A modified spectral normalization that preserves histological information.	• Partially preserves the color variation and structural information present in source image. • Solves the problem of inconsistencies in biopsy imaging and color variability.	• Having limitations in immunohistochemical stains that do not satisfy the Beer-Lamber law. • This method is not applicable before stains are completely independent to each other. • High computational complexity.

(*continued on next page*)

Table A.1 (*continued*).

Methods	Descriptions	Advantages	Limitations
Hoque et al. [9]	• Retinex model based stain color normalization method. • A group of pixels is separated and converted into optical density space. • Normalized hematoxylin and eosin components are obtained with a multiscale Retinex model that estimates the illumination map of hematoxylin and eosin components. • Estimates illumination factors individually by finding the maximum intensity of pixels in three color channels. • Refines illumination map by exploiting the structure of illumination map. • Manipulates illumination map with the help of Gamma correction in the range of [0, 1]. • Finally, recombines hematoxylin and eosin components as output of normalization result.	• Removes the color artifacts and effectively estimates the hematoxylin, eosin, and background distributions. • Classifies the pixels with better performance in color distribution of the monochromatic whole slide image. • Having low computational cost and easy to use in effective computer aid systems. • Only estimates the illumination map of the whole slide image and utilizes more general color pixels without any apparent color discontinuity. So, the normalized images are consistent in color appearance with less color artifacts.	• The preliminary stain separation might be failed if one of two stains is not present in the image properly. • It is hard to get back the original color, once the normalization process is performed.
Liang et al. [55]	• An approach that includes two stain style transfer models. • Performs stain transformation using GAN and directional statistics-based color similarity index as loss function for the preservation of texture, structure, and chroma-contrast features. • A pre-trained classifier network is used to preserve features and learn color distributions. • A fully 1×1 convolutional network is used for stain transformation and another network is trained based on the method in [51] to generate paired images.	• The texture, structure, and chroma-contrast features are preserved in normalized image. • A noticeable normalization performance in regions where stains mix due to co-localization of histological substances.	• Color information may act as noise and introduce overfitting problems. • High computational complexity. • Longer estimation time is required.
Moghadam et al. [14]	• Two models are developed for stain transformation with conditional GANs. • First model has one network per stain style transformation. • Second model uses only one network for many-to-many stain style transformations.	• Performs many-to-many stain transformation without source domain labels. • The models extract color-related and structural features with neural networks which helps to extract features based on many-to-one stain transformations. • Solves the problems of requiring pre-processing, paired labels and reliance on reference image.	• This approach is not proficient to maintain the structural contents in histological images. • High optimization costs. • Longer training time is required due to the two-to-two stain transformations which might be attributed to the higher number of loss functions.
Kausar et al. [89]	• A deep model, called stain acclimation generative adversarial network (SA-GAN), includes one generator and two discriminators. • Color transformation is performed considering the image-specific color patterns as style patterns. • The generator helps to map the inputs from source domain to target domain. • The color patterns as of target domain is maintained and the structure contents as of source domain is preserved by two discriminators respectively. • The color attribute constraint metric extracted from the template image is used to train the model.	• This model learns both the dataset-specific staining properties and image-specific textural contents. • Performs an efficient mapping between data domains by involving entire domain distributions. • The tissue structures of images are not altered and the structure of original image is preserved.	• Considering color patterns of images as style patterns and carrying out color transformation creates unwanted artifacts. • Suffers from the limitation of generating samples with little diversity. • Suffers from high computational cost and time complexity.

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